

ABC INTERNATIONAL SCHOOL

Powai, Mumbai – Maharashtra | IB & CBSE

LABORATORY OPERATIONS & SAFETY MANUAL

Science Department Dataset & Compliance Guide

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Field	Details
School	ABC International School – Powai Campus
Location	ABC International School, Central Avenue, Hiranandani Gardens, Powai, Mumbai – 400 076
Board	IB (PYP/MYP/DP) & CBSE
Laboratories	Physics, Chemistry, Biology, Junior Science Lab
Compliance	IBO Health & Safety Regulations
Primary Focus	Student Safety, Equipment Integrity, Chemical Compliance
Staff Strength	12 Science Faculty
Document Scope	Safety SOPs, Usage Rules, Inventory, Emergency Protocols
Prepared By	Science Department & Safety Committee
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1. LABORATORY SAFETY SOP

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1.1 Purpose & Scope

This SOP defines the safety protocols for all science laboratories at ABC International School. It is mandatory for all students, faculty, and lab assistants to adhere to these guidelines to prevent accidents and ensure a safe learning environment.

1.2 Personal Protective Equipment (PPE) Matrix

Lab Type	Safety Goggles	Lab Coat	Gloves (Nitrile)	Closed-Toe Shoes
Chemistry	Mandatory	Mandatory	Mandatory	Mandatory
Biology	Mandatory	Mandatory	Mandatory	Mandatory
Physics	Optional*	Optional	Optional	Mandatory
Junior Lab	Mandatory	Mandatory	Optional	Mandatory

**Mandatory during optics or electrical experiments involving high voltage.*

1.3 Pre-Experiment Safety Checklist

#	Item	Verification	Responsibility
1	Fume Hood Functionality	Check Airflow	Lab Assistant
2	Eyewash Station	Flush for 30s	Lab Assistant
3	Fire Extinguisher	Pressure Check	Safety Officer
4	First Aid Kit	Stock Verification	Lab Assistant
5	Gas Leak Detection	Manual Sniff Test	Lab Assistant

2. GENERAL LABORATORY USAGE RULES

2.1 Student Conduct

- Supervision:** No student is allowed in the lab without a teacher or lab assistant present.

2. **Food/Drink:** Strictly prohibited in all laboratory areas to prevent contamination/ingestion.
3. **Behavior:** No running, practical jokes, or unauthorized experiments.
4. **Housekeeping:** Work areas must be cleaned and apparatus returned to designated spots after every session.

2.2 Lab Booking & Access

Activity	Process	Timeline
Regular Class	Scheduled in Timetable	Weekly
Extra Session	Request via Admin Portal	48 Hours Notice
Project Work	Permission from HOD	1 Week Notice

3. CHEMICAL INVENTORY & STORAGE

3.1 Hazardous Material Classification (GHS)

Category	Storage Requirement	Example
Flammable	Fire-rated Metal Cabinet	Ethanol, Acetone
Corrosive	Acid-resistant Poly Cabinet	HCl, H ₂ SO ₄
Oxidizer	Separate from Flammables	Potassium Permanganate
Toxic	Locked Cabinet	Barium Chloride

3.2 Sample Inventory Log (Chemistry Lab)

Chemical Name	Grade	Quantity	Expiry Date	Location
Sodium Hydroxide	AR	2 kg	Dec 2027	Shelf C-04
Silver Nitrate	LR	250 g	Jan 2028	Locked Box 01
Phenolphthalein	LR	500 ml	Mar 2027	Cabinet D-02

4. EQUIPMENT MAINTENANCE & CALIBRATION

4.1 Maintenance Schedule

Equipment	Frequency	Activity	Responsible
Electronic Balance	Monthly	Calibration with Standard Weights	Lab Assistant
Microscopes	Quarterly	Lens Cleaning & Service	External Vendor
Fume Hoods	Bi-Annual	Face Velocity Testing	Facilities Dept
Autoclave	Monthly	Pressure & Seal Check	Lab Assistant

5. LABORATORY EMERGENCY & SPILL PROTOCOLS

5.1 Emergency Response Matrix

Incident	Immediate Action	Contact
Chemical Splash (Eye)	Flush at Eyewash for 15 mins	Medical Room
Chemical Spill (Floor)	Contain with Spill Kit (Sand/Neutralizer)	Lab Assistant
Fire (Small)	Use Fire Blanket / CO2 Extinguisher	Safety Officer
Fire (Large)	Sound Alarm & Evacuate	Fire Dept (101)

5.2 Spill Kit Contents

- Neutralizing agents (Sodium Bicarbonate for acids, Citric Acid for bases).
 - Absorbent pads and socks.
 - Disposal bags and zip ties.
 - Dustpan and brush (non-sparking).
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6. HAZARDOUS WASTE DISPOSAL

6.1 Waste Categorization

- **Blue Bin:** Glassware (broken or contaminated).
- **Yellow Bin:** Biological waste (agar plates, dissected tissues).
- **Red Bin:** Sharp objects (needles, scalpels).

- **Chemical Waste:** Liquid waste must be neutralized before disposal or collected in designated 5L carboys for professional pickup.
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7. STUDENT SAFETY CONTRACT (EXCERPT)

"I, [Student Name], understand that the science laboratory is a place of serious work. I agree to follow all safety instructions provided by my teacher. I understand that failure to comply with safety rules may result in my removal from the laboratory and further disciplinary action."

Student Signature: _____ Date: _____ Parent Signature: _____
Date: _____

8. LAB INCIDENT KNOWLEDGE BASE

Q1: What should I do if I break a mercury thermometer? A: Do not touch the mercury. Evacuate the immediate area. Inform the lab assistant who will use the 'Mercury Spill Kit' to collect the droplets for hazardous disposal.

Q2: How do I request new chemicals for a student project? A: Submit a 'Lab Requisition Form' to the HOD at least 2 weeks in advance. Ensure the MSDS (Material Safety Data Sheet) is reviewed before the experiment begins.

Q3: Are contact lenses allowed in the Chemistry lab? A: It is strongly recommended to wear prescription glasses instead of contact lenses, as lenses can trap chemical vapors against the eye. If worn, non-vented safety goggles are mandatory.